



EE 208: Digital Electronic Circuits

Course Code	EE 208
Course Name	Digital Electronic Circuits
L-T-P-C	3-0-0-3
Intended for	B.Tech.
One-time approval	8th Senate (Oct 2015)
Recoded to	EE 210 by the 30th Senate (Feb 2021)

Course Contents

- **Number system:** Binary Numbers, Octal Numbers, Hexadecimal Numbers, Complement, Signed Binary Numbers Addition and Subtraction
- **Logic Gates and Boolean Algebra:** Digital Logic Gates, Basic Theorems and Properties of Boolean algebra, Boolean Functions, Minterms and Maxterms, Sum of Products and Product of Sums
- **Minimization of Logic functions:**
- Karnaugh map method, Simplification of logic expressions, two variable, three variable, four variable etc., Implementation of logic functions
- **Combinational Circuits :** Design of Adders, Subtractor, Multiplier, Encoder & Decoder, Multiplexer & Demultiplexer, and their use in logic synthesis, Arithmetic circuits, Seven-segment and alphanumeric display design.
- **Sequential Circuits:** Latch, RS, JK, Master Slave, D, and T flip flops Finite state machines, State reduction
- **Registers and Counters:** Registers, shift registers, Counters, Asynchronous Counters, Synchronous Counters, Design of counters, design of other sequential circuits.
- **Memory and Programmable Logic devices**
- **Digital Integrated Circuits**

Text Books

1. Mano, M.M. and Ciletti, M.D., **Digital Design**, 4th Edition, Prentice-Hall.
2. Floyd, T.L., **Digital Fundamentals**, 8th Edition, Pearson Education.

Other Textbooks

1. Balabanian, N. and Carlson, B., **Digital Logic Design Principles**, John Wiley & Sons.
2. Jain, R.P., **Modern Digital Electronics**, 3rd Ed., Tata McGraw-Hill.
3. John F. Wakerly, **Digital Design Principles and Practices**, Prentice Hall.